

MLA 0202 - Fundamentals of Machine Learning

Assessment Tool 3

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Experiment 1: Probability Theory

Aim

To calculate the prior probability of Malignant and Benign classes using the Breast Cancer Wisconsin Diagnostic Dataset and predict the class of a new instance.

Step-by-Step Discussion

1. Load the Breast Cancer Wisconsin Diagnostic Dataset.
2. Separate feature values and target labels.
3. Count the Malignant and Benign samples.
4. Calculate prior probabilities for both classes.
5. Display the probabilities.
6. Predict the class of a new instance using the higher prior probability.

Output

Total Samples: 569 Malignant: 212 Benign: 357 $P(\text{Malignant})=0.3726$
 $P(\text{Benign})=0.6274$ Predicted Class: Benign

Result

The probabilities of both classes were computed successfully. The new instance was predicted as Benign because it has the higher prior probability.

Result Analysis

This experiment demonstrates how probability theory estimates the likelihood of each class from the dataset. These probabilities can be used as prior knowledge for prediction.

Experiment 2: Bayes Decision Theory

Aim

To classify SMS messages as Spam or Ham using the SMS Spam Collection Dataset and Naive Bayes.

Step-by-Step Discussion

1. Load the SMS Spam Collection Dataset.
2. Split the dataset into training and testing data.
3. Convert messages into numerical vectors.
4. Train the Multinomial Naive Bayes classifier.
5. Calculate posterior probabilities.
6. Predict whether the given message is Spam or Ham.

Output

Total Messages: 5572 Model Accuracy: ~98% Posterior Probability: Ham : 0.0000 Spam: 1.0000 Prediction: spam

Result

The classifier successfully identified the given message as Spam after calculating posterior probabilities.

Result Analysis

Bayes Decision Theory uses prior and conditional probabilities to estimate posterior probabilities. Naive Bayes performs efficiently for text classification and provides high prediction accuracy on SMS spam detection.

Experiment 3: Information Theory

Aim

To calculate the entropy and information gain of all attributes in the Play Tennis dataset and identify the best splitting attribute.

Step-by-Step Discussion

1. Create the Play Tennis dataset.
2. Calculate the entropy of the target attribute.
3. Calculate the information gain for each input attribute.
4. Compare the information gain values.
5. Identify the attribute with the highest information gain.

Output

Entropy = 0.9403 Information Gain Outlook = 0.2467 Temperature = 0.0292
Humidity = 0.1518 Wind = 0.0481 Highest Information Gain = Outlook

Result

The entropy and information gain were calculated successfully. Outlook produced the highest information gain and is selected as the root attribute.

Result Analysis

Information gain measures the reduction in uncertainty after splitting the dataset. The attribute with the highest value is considered the best choice for building a decision tree.